

The Flow Clock

Trading the Treasury market's calendar of forced flows: auction supply and month-end index demand

Gator Quant Hacks 2026 · Systematic Trading track · Team: Siddharth Radhakrishnan, Siddhant Pallod, Atharv Diwan, Imbisat Malik · Code and data: github.com/sidrad17/flow-clock (one command: `python run_all.py`, no API key) · Hypotheses timestamped in git before testing · In-sample 1993-01 to 2024-09; test window 2024-10 to 2026-09, run once

Summary

- **Two groups must trade Treasuries on dates known in advance:** dealers who absorb new bonds at coupon auctions, and bond index funds that rebalance at month-end. Their forced trades move prices for a few days. The Flow Clock trades that calendar. It never forecasts the level of rates.
- **The trade we call tradable held up out of sample.** Long 10-year note futures (ZN) over the last four days of each month: net Sharpe 0.79 in-sample (2010-07 to 2024-09; 0.71 at 2× costs, worst drawdown 6.7%, capacity ≈ \$1.41B) and **0.64 in the untouched test window** (2024-10 to 2026-09; 0.55 at 2× costs, $t = 0.91$, worst drawdown 4.1%), run once on frozen code. That confirms the sign; 24 months cannot establish significance.
- **In-sample, the textbook explanation failed and the auction cycle explained part of the rally.** The rally is real (+0.19% per month-end, $t = 4.48$), but index funds' forced duration demand (FDD), rebuilt point-in-time since 1993, did not predict its size ($b = 0.011$, $t = 0.17$). Treasuries dip before coupon auctions and bounce after ($t = -3.31 / 3.44$), more after larger auctions ($t = 3.43$), and the bounce after late-month auctions explained part of the rally (H7, $t = 2.35$).
- **Out of sample, both explanations flipped.** FDD's slope had the predicted sign ($b = 0.88$, $t = 2.99$; forecast-sized 1.20 vs calendar-only 0.71), and H7 reversed ($t = -2.21$). The cash auction leg lost money, and the Flow Clock book (−0.18) met its pre-registered kill condition. With 24 test months against 381 in-sample months, we report these as the rules label them and treat the *mechanism* as unresolved.
- **What we claim:** a pre-registered, reproducible month-end trade in ZN futures that survived its one-shot test. **What we do not claim:** that the auction trade is tradable (it failed our kill tests), that dealer inventory drives it (H8 failed twice), or that we know which forced flow drives the month-end rally.

1. Economic foundation

Edge source: structural constraint plus liquidity provision. Two groups trade on fixed dates for reasons unrelated to value. *Primary dealers* must take down new supply at each coupon auction. They hedge it in advance and need several days to distribute it. Their balance sheets are limited, so the market pays them a concession: yields rise into the auction and fall after it. *Index-tracking bond funds* are judged on matching their index, not on when they trade. At month-end the index adds new issues and reinvests the month's coupon cash, so trackers must buy duration in the last days of the month.

Who is on the other side. Before an auction we sell alongside dealers' hedging. After it we buy the inventory they need to place, and at month-end we pre-position for trackers' demand. In both cases we are paid for immediacy. **Why the edge persists.** The calendar is public, but the forced players cannot move their dates. Warehousing duration around these dates carries rate risk several times the size of the edge. Capacity is about \$1.3–1.4B (section 7), too small for large balance sheets to arbitrage away. The NY Fed finds the auction pattern still present in recent data (Fleming, Liu & Nguyen, 2026).

What is new. (1) A point-in-time rebuild of the Treasury index's forced duration demand from public auction records, with Fed SOMA holdings deducted by CUSIP, and a clean rejection of the forced-demand explanation. (2) A test of the link between the two clocks: late-month auction supply explained part of the month-end rally in-sample (H7), though not in the test window. (3) One DV01-netted book that trades both sides, a test of whether it survives in futures, and a diagnostic showing that the cash auction effect is not an artifact of how the constant-maturity curve is built. (4) Every hypothesis was committed and tagged in git before its first return was computed, with a logged trial count and a test window run once.

Tag (commit)	Committed (ET)	What it fixed before any related result
gate1-prereg (745354e)	Oct 3, 1:00 AM	Month-end hypothesis H1–H5, every parameter (HYPOTHESIS.md, config/settings.py; unchanged since), before any data download
prereg-addendum (dbfe85e)	Oct 3, 3:34 AM	SOMA deduction by CUSIP; refunding-month analyses reported beside H1
prereg-flowclock (8c41154)	Oct 3, 4:27 AM	Auction hypothesis H6a–c, H7, the Flow Clock book and its kill conditions, before any auction-window return
prereg-dealers (15f67bc)	Oct 3, 10:27 PM	H8 dealer-inventory test, before any dealer-position data was loaded
gate2-frozen (825135d)	Oct 4, 1:15 AM	Code frozen; test-window rules (written first, f2d3531) and the one-time <code>--oos</code> run

2. Data

Source	Series	Use
US Treasury Fiscal Data (no key)	Every coupon auction since 1979: dates, sizes, CUSIPs	Auction events, size signal, index rebuild
FRED (no key)	Constant-maturity yields 1–30 years; 3-month T-bill	Cash returns, DV01, financing, bond calendar
NY Fed Markets Data	SOMA holdings by CUSIP; weekly primary-dealer positions	Fed-held bonds removed from the index; H8
Ken French; Federal Reserve	Daily US equity return; scheduled FOMC dates	Pension-rebalancing control (H5); FOMC risk rule
Databento CME (GLBX.MDP3, licensed)	Settlements, volume, definitions: ZT, ZF, ZN, TN, ZB, UB, from 2010-06	Futures layer only; only derived tables are committed

Point in time. Every signal uses data dated before entry. Auction sizes count only if announced by A–5. z-scores use past values only. The FOMC rule uses scheduled dates only. Public data is frozen in the repo with SHA-256 checksums, and every in-sample loader cuts at 2024-09-30. **Returns.** Cash returns are excess returns over T-bills of a constant-maturity par bond, computed from FRED yields. That curve is fitted, so cash results are not directly tradable, which is why we also test futures. Futures P&L is settlement to settlement, holding the contract whose first intention day is more than 5 business days after exit.

3. Model

No machine learning: about 380 months is too few to fit one without overfitting. Every rule is fixed in a file with a timestamp and can be read in the code.

Signals. Auction size relative to recent auctions of the same maturity, known at the close of A–5:

$$zS_e = (S_e - \text{mean of previous 6}) / \text{sd of previous 6, clipped to } \pm 3$$

Late-month supply: zA_m = standardized $\Sigma(\text{size} \times \text{duration})$ of auctions held from T–8 to T–4. Forced demand (the failed signal): $FDD_m = \text{Ext}_m + c_m \times D_{\text{next}}$, with z_m its past-only z-score and forecast weights $w_m = \text{clip}(1 + z_m, 0, 2)$.

Legs. *Supply:* short the auctioned maturity from close A–5 to close A, long from A to A+5. Each leg is sized in DV01 so that a 1-sd 5-day move costs 0.25% of capital (60-day vol). *Demand:* long the 10-year from T–4 to T, sized so that a 1-sd 4-day move costs 1%. *Book:* positions netted in DV01 by maturity each day, costs on net DV01 traded. *Futures map:* 2y/3y → ZT, 5y → ZF, 7y → ZN, 10y → TN (ZN before 2016), 20y → ZB, 30y → UB; month-end → ZN.

Costs. Cash: 0.5bp of yield per round trip, 1.4–1.8× the interdealer spreads in Fleming (2003). Futures: 1 tick + \$2 per contract per round trip. Every result is also shown at 2× costs. **Inference.** Monthly regressions use Newey-West errors. Auction tests cluster by calendar week (auctions in one week share rate shocks). Sharpe differences use a paired bootstrap by month (10,000 draws).

4. Results

Scorecard of pre-registered tests (in-sample 1993-01 to 2024-09; futures 2010-07 to 2024-09; returns in %). The test-window column applies the labels written before the run: H1 can only "confirm the sign" (24 months give $t \approx 1.4$ even for a true Sharpe of 1), H4 and H8 are pass/fail, and everything else is "consistent" if its sign matches in-sample.

Test	Prediction	In-sample result	Verdict	Test window
Month-end rally (10-year, T–4 to T)	mean > 0	+0.194% per month-end, $t = 4.48$; above all 1,000 random 4-day windows	Passed	+0.151 consistent
H1: index funds' forced demand (FDD) sets its size	b > 0	b = 0.011, $t = 0.17$ (n = 381)	Failed	b = 0.875, $t = 2.99$ (n = 24) sign confirmed
H4: forecast-sized beats calendar-only	Sharpe diff > 0	0.57 vs 0.63 (diff –0.05)	Failed	1.20 vs 0.71 (diff +0.49; p = 0.017) pass
H3: partial reversal after month-end	mean R3 < 0, slope < 0	R3 –0.063% ($t = -1.23$); slope +0.079	Not supported	–0.023 ($t = -0.21$) consistent
Placebo: business days 4–7	no rally	–0.042% ($t = -0.90$)	As predicted	+0.000 ($t = 0.00$) not consistent
H6a: dip before coupon auctions (A–5 to A)	mean < 0	–0.102%, $t = -3.31$ (n = 1,694)	Passed	–0.137 ($t = -1.37$) consistent
H6b: bounce after (A to A+5)	mean > 0	+0.114%, $t = 3.44$	Passed	–0.057 ($t = -0.74$) not consistent
H6c: larger auctions, larger swing	$\beta > 0$	$\beta = 0.135$, $t = 3.43$	Passed	–0.161 ($t = -0.75$) not consistent
H7: late-month auction supply explains part of the month-end rally	c > 0, a > 0	c = 0.056 ($t = 2.35$); a = 0.162 ($t = 3.49$)	Passed	–0.246 ($t = -2.21$) not consistent
Headline 3: book beats demand leg alone, net	Sharpe diff > 0	0.87 vs 0.63; p = 0.06	Passed	–0.90 [–2.28, 0.45] not consistent
Auction leg survives costs in futures	kill if it disappears at 2×	net Sharpe 0.29 at 1×, 0.07 at 2×	Failed	0.60 at 1×, 0.37 at 2× consistent

Test	Prediction	In-sample result	Verdict	Test window
H8: dealer inventory drives the auction effect	$c > 0, p < 0.05$	$c = 0.006, t = 0.15, p = 0.44$ (n = 1,191)	Failed	$c = -0.021, p = 0.57$ (n = 101) fail

Strategies (net of costs; max drawdown of the excess-return NAV; "Curve" = cash constant-maturity, not directly tradable):

Strategy	In-sample	Sharpe 1×	Sharpe 2×	Max DD	Tradable?	Test 1×	Test 2×
Month-end, ZN futures	2010-07-2024-09	0.79	0.71	6.7%	Yes	0.64	0.55
Month-end, cash, calendar-only	1993-01-2024-09	0.63	0.45	8.0%	Curve	0.71	0.51
Month-end, cash, forecast-sized (FDD)	1993-01-2024-09	0.57	0.42	10.7%	Curve	1.20	1.02
Month-end, cash, curve-allocated	1993-01-2024-09	0.69	0.53	7.9%	Curve	0.73	0.52
Auction supply leg, cash	1993-01-2024-09	0.80	0.46	9.5%	Curve	-0.68	-1.11
Flow Clock book, cash	1993-01-2024-09	0.87	0.51	11.1%	Curve	-0.18	-0.55
Auction supply leg, futures	2010-07-2024-09	0.29	0.07	6.5%	Fails kill test	0.60	0.37

Treasuries around coupon auctions, by auction size vs the previous six (1993-01 to 2024-09)

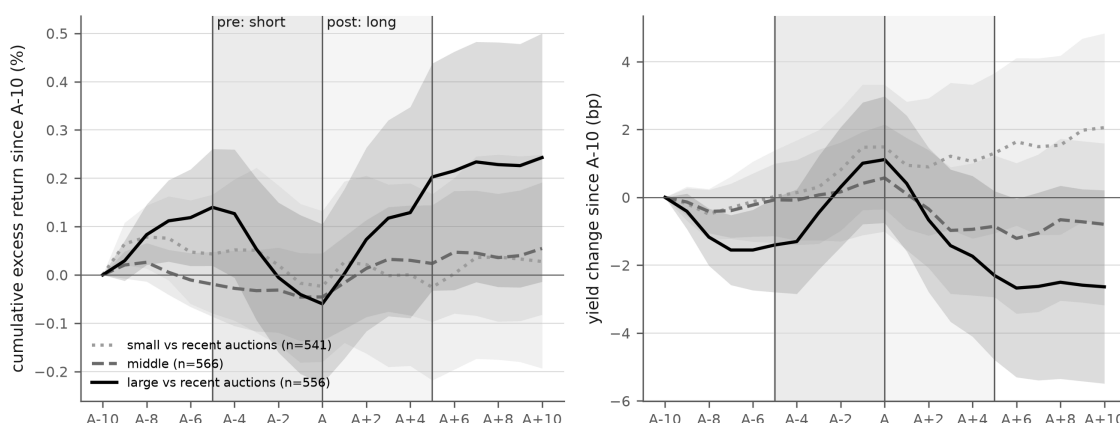


Figure 1. Cumulative excess return (left) and yield change (right) around coupon auctions, by auction size against the previous six of the same maturity, with 95% bands clustered by week. Large auctions: -0.202% from A-5 to A, then $+0.263\%$ from A to A+5. Small auctions barely move.

Reading the results. In-sample, the auction effect is the strongest result. Across all events, post-minus-pre averages $+0.216\%$ per auction ($t = 5.07, n = 1,694$). It holds after the original study's sample, 2009–2024 (pre $t = -3.10$, post $t = 1.96$, size $t = 2.34$; book Sharpe 0.98). It is **weak in 2020–24**: the book's Sharpe by period is 1993–1999: 0.52 · 2000–2009: 1.25 · 2010–2019: 1.15 · 2020–2024: 0.26, and the auction long-short there has $t = 0.18$. The 20-year bond ($n = 54$) and the 7-year's pre-auction window go the wrong way. The book beats the demand leg alone by $+0.24$ Sharpe, but the bootstrap interval includes zero ($p = 0.06$), and at $2\times$ costs the gap is $+0.06$.

Cash vs futures. The month-end trade carries over: ZN futures and the cash trade on the same days correlate 0.95 daily, with Sharpe 0.79 vs 0.71. The auction leg does not. At equal risk, futures capture 69% of the cash move before costs, and higher turnover takes the Sharpe from 0.54 gross to 0.29 net. **Is the cash effect an artifact of the fitted curve?** We checked, and it is not. 89% of the cash-minus-futures gap falls outside auction and settlement days, and 54% falls before the auction. Reopenings, where the curve's input bond does not change, show the effect too ($+0.31\%$, $t = 2.18, n = 338$). Why futures capture less remains open.

5. Test window: 2024-10 to 2026-09, run once

The code was frozen and tagged `gate2-frozen` (825135d). The rules for labelling every test-window number were committed before the code (f2d3531). `python run_all.py --oos` then ran once at 06:00 UTC on Oct 4. It downloaded the public and futures test-window data fresh and logged 14 trial rows. Nothing was re-run or changed.

Month-end leg: consistent. The rally averaged $+0.15\%$ per month-end ($t = 0.84, n = 24$). Net Sharpe was 0.64 in ZN futures and 0.71 in cash. HYPOTHESIS.md's kill conditions were not met: $b > 0$, reversal $R3 = -0.023\%$, and forecast-sized beat calendar-only.

FDD: sign confirmed, H4 pass, under the pre-registered labels. This is the opposite of in-sample. Inside the window the evidence is mixed. The two FDD components point the other way from in-sample. The cross-maturity test H2 has the wrong sign ($-2.49, t = -1.92$). The surprise component is strongly negative ($t = -4.54$). In the placebo window, sizing by FDD also "wins" ($+0.37$, interval includes zero). We report the labels as written. We do not call FDD a success on 24 months, just as we would not have dropped it on 24.

Auction leg: not consistent. Across 167 auctions, the pre-auction dip kept its sign ($t = -1.37$), but the post-auction bounce and the size effect did not ($t = -0.74, t = -0.75$). The cash supply leg earned -0.68 . The Flow Clock book earned -0.18 against 0.71 for the demand leg alone, so PREREG_FLOWCLOCK's kill condition is met. The futures auction leg earned 0.60 (0.37 at $2\times$), the reverse of in-sample. We keep its in-sample verdict. H8 failed again ($c = -0.021, p = 0.57, n = 101$).

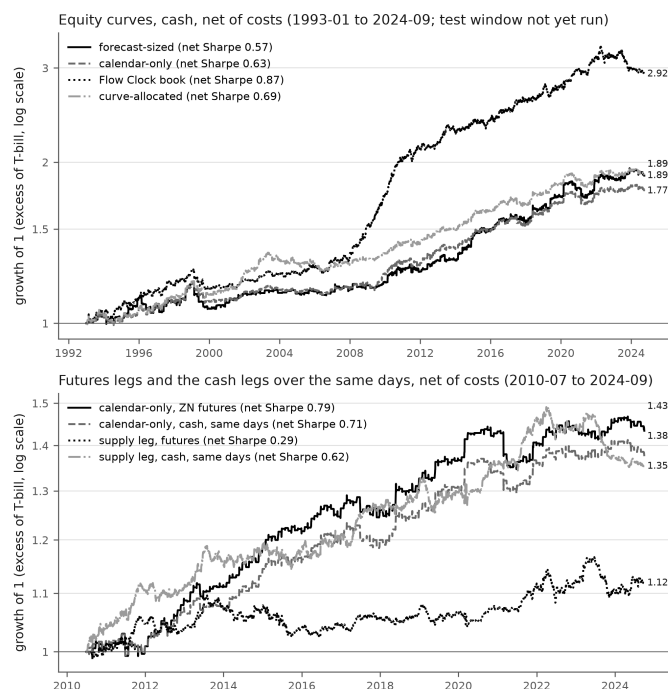


Figure 2. In-sample: growth of 1 in excess of T-bills, net of costs, log scale. Top: cash strategies, 1993–2024. Bottom: futures legs and the cash legs on the same days, 2010–2024.

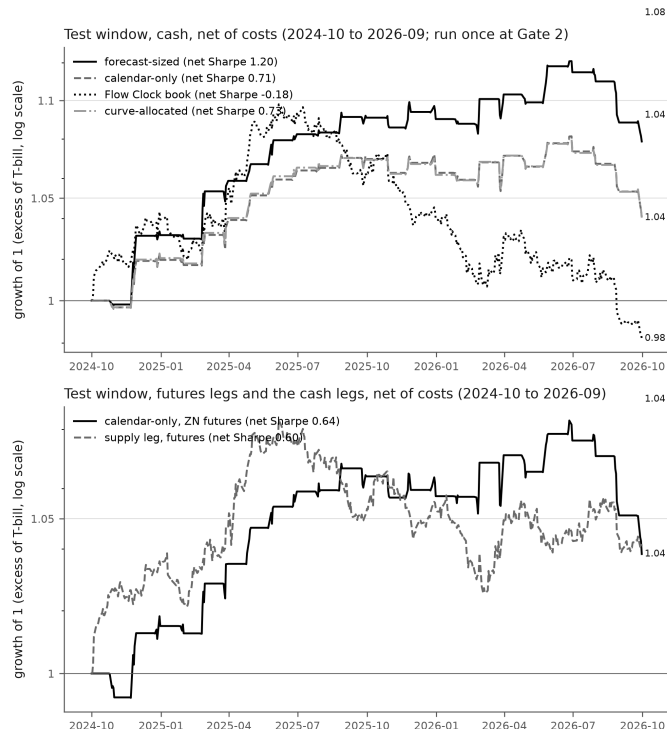


Figure 3. Test window, 2024-10 to 2026-09, same construction, run once (outputs/figures/equity_curve_oos.png). Top: cash strategies. Bottom: ZN month-end and the futures auction leg.

6. Risk management

Rules (all pre-registered, each with an on/off test). (1) Volatility-targeted DV01 sizing from 60-day volatility. (2) Half size when a scheduled FOMC decision falls inside the holding window; unscheduled meetings are never used. (3) Drawdown rule: when the strategy's own drawdown exceeds $2\times$ its expected yearly volatility, half size until a new high. (4) Gross notional capped at $3\times$ capital.

The rules cut the worst drawdown, at a small cost in Sharpe; the cap costs most. **Tail risk is a rates shock inside the window.** The worst month-end loss is -3.4% in 2011-06, when the 10-year sold off 30bp in four days (3.5 sd). The book's worst year is 1999. **Exposures.** The month-end leg is long duration a few days a month (10-year beta 0.15). The supply leg has no bond beta (0.003), so its alpha stands alone (2.4% a year, $t = 4.52$). Equity beta is about zero throughout. The two legs' daily returns correlate 0.20.

Cash, 1993–2024	Book	DD	Month-end	DD
All rules on (as traded)	0.87	11.1%	0.63	8.0%
FOMC half size off	0.84	11.2%	0.62	7.3%
Drawdown rule off	0.85	12.9%	0.63	8.0%
$3\times$ notional cap off	0.92	11.1%	0.63	8.0%
All off	0.90	12.3%	0.61	9.9%

7. Liquidity and capital

Capacity uses square-root market impact on average daily volume (ADV), with each leg capped at 5% of ADV. "Halves at" is the capital at which net Sharpe falls to half.

Leg	Volume measure	Sharpe at \$10M	Halves at	Note
Month-end, ZN futures	Most-active contract	0.75	\$1.41B	Median ADV 1.23M contracts
Month-end, ZN futures	Contract held (as first written)	0.59	\$333M	Roll months read the thin new contract; both reported
Month-end, cash	Primary-dealer volume	0.69	\$1.33B	Median ADV \$93B
Auction leg, futures	Most-active contract	0.21	\$60M	Fails its kill test anyway

The ZN month-end leg turns over $37\times$ a year and pays 0.26% a year in costs. It uses $1.6\times$ capital in notional on average while open, and the $3\times$ cap bound on 2 of 169 months.

8. What failed, and what it tells us

- Forced index demand (H1, H4, H3).** In-sample, index funds' forced buying did not set the size of the rally, sizing by it did not help, and the predicted reversal was absent. In 0 of 480 sensitivity cells did H1's 95% interval exclude zero. The test window gave the opposite answer, on

24 months and with mixed internal evidence (section 5). We drop it as a *trading* signal, since calendar-only is simpler and did well in both samples, and we leave it open as an *explanation*.

- **The auction leg.** It is strong in cash in-sample and not a curve artifact. But it fails the futures kill condition in-sample, its cash version lost money in the test window, H7 reversed there, and 2020–24 was already its weakest period. Either the effect has faded since dealers' role changed, or it was never large enough to trade. We do not claim it is tradable.
- **Dealer inventory (H8), failed twice.** Weekly dealer positions do not predict the auction swing ($c = 0.006$, $t = 0.15$ in-sample), while auction size does in the same regression ($t = 2.66$). It is the flow, not the stock. Caveats: the data are weekly, cover all maturities together, and dealers hedge in futures.

9. Honesty box

- **Trials.** 519 distinct variants were tested and logged (3,115 runs in `runs/trials.csv`). The Deflated Sharpe at $N = 519$ is 0.987 for the book, 0.874 for ZN month-end and 0.236 for the futures auction leg. For the book, the chance of a Sharpe this high from luck alone, after 519 tries, is about 1.3%. Counting every logged run ($N = 3,115$): 0.957 and 0.793.
- **Robustness.** Cash calendar-only is profitable in 100% of 480 sensitivity cells (entry, exit, index-rebuild variants). The placebo window (business days 4–7) shows no rally. The month-end mean beats all 1,000 random 4-day windows (their 97.5th percentile: 0.062%).
- **Known flaws, disclosed and not fixed after seeing results.** 66 of 662 2-year futures auction legs, in the zero-rate years (2011–14, 2020–21), got a weak DV01 fit ($R^2 < 0.5$), and 62 hit the 3× cap; the month-end leg is unaffected. In 1 of about 95 full runs, 24 of the 480 sensitivity cells (one rebuild variant) came out different, for an unknown reason; the headline numbers never moved. Cash costs (0.5bp) are our assumption, checked against published spreads.
- **Data hygiene.** The public snapshot downloaded on Oct 3 contains rows after 2024-09-30. Every loader cut at 2024-09-30, and a date guard refused later dates until the `gate2-frozen` tag existed. The public repo carries the full history except one force-pushed commit that held licensed futures data. The original repo's activity log is available on request.
- **Test window.** Run once at 825135d (06:00 UTC, Oct 4); 14 trial rows logged; no rerun. Databento flagged a few test-window days as reduced quality (for example 2025-09-17, 2025-09-24 and 2025-11-28). We used them as delivered.
- **Reproduce.** `python run_all.py` rebuilds every in-sample number from the committed public data in about 2 minutes, with no key. A fresh clone matched exactly on macOS and on Linux (0 of 19,218 values differ).

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